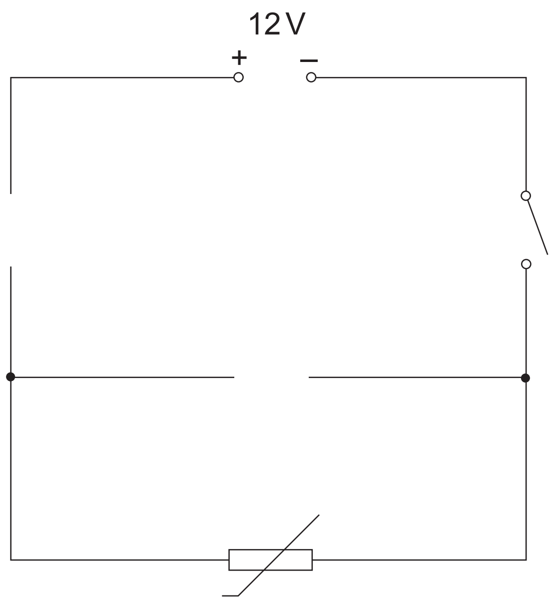
Examiner only

5. Tommy and Arthur are investigating the effect of temperature on the resistance of a thermistor.

The circuit diagram drawn by Tommy is shown below.



(a) Complete the circuit above by adding:

(i) an **ammeter** to measure current through the thermistor. [1]

(ii) a **voltmeter** to measure voltage across the thermistor. [1]

(b) The results of Tommy's experiment are shown in the table below.

Temperature (°C)	Current (A)	Voltage (V)	Resistance (Ω)
0	0.05	12	240
20	0.10	12	120
40	0.20	12	60
60	0.40	12	30
80	0.80	12	15
90	1.1	12	

Use only the information in the table to answer the following questions.

(i) State the independent variable in this experiment. [1]

....

(ii) State a controlled variable in this experiment. [1]

....

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(c) Use the information in the table to answer the following questions.

(i) Estimate the **current** when the temperature is 100 °C. [2]

current = ... A

(ii) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the thermistor at 90 °C. [2]

resistance = ... Ω

(iii) Tommy thinks that when he increases the temperature by 20 °C the resistance of the thermistor decreases by half every time. Use Tommy's results to explain whether you agree. [2]

....

....

....

(iv) Arthur wants to know if the thermistor they used in the circuit would be suitable as a temperature sensor. The sensor must work between 20 °C to 60 °C and vary in resistance by at least 100 Ω.
Use Tommy's results to explain whether the thermistor would be suitable. [2]

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